SFI

Self-Focus Index (SFI) Notebook — README

Purpose

The **Self-Focus Index (SFI)** measures how much a model's final generated token attends to its own recent output versus the full reasoning context.
- **High SFI → more self-referential attention → higher hallucination risk**
- **Low SFI → grounded, stable reasoning → better hallucination mitigation**

Because each mitigation method (ToT, RAG, KG, CAD) generates attention matrix (AMA) files, this notebook computes SFI across all 338 examples for each method and compares their overall stability.

Required Libraries

If running locally or outside Colab, install:

```
pip install torch numpy pandas matplotlib
```

Python version **3.10+** recommended.

No language model is loaded—this notebook processes attention matrices only.

Required Input Files

You will receive only the notebook, so you must upload your AMA folders into your own Google Drive.

Method	Folder YOU must upload	Contents
ToT	`/YOUR_FOLDER/AMA/`	338 `attn_XXXX.json` files
RAG	`/YOUR_FOLDER/RAG_attention_matrices/`	338 JSON files
KG	`/YOUR_FOLDER/kg_attention/`	338 JSON files
CAD	`/YOUR_FOLDER/cad_attn/`	338 JSON files

After uploading, update the notebook paths such as:

```
AMA_DIR = "/content/drive/MyDrive/YOUR_FOLDER/AMA"
```

How to Run the Notebook (Colab Recommended)

1. Open the notebook in **Google Colab**.
2. Upload the AMA folders to your Google Drive.
3. Update all AMA paths in the notebook to match your Drive.
4. Mount Drive using the provided cell.
5. Run the notebook **top to bottom**.

The notebook will: - Load all AMA JSON files
- Compute SFI for each example
- Display SFI samples for verification
- Compute average SFI for ToT, RAG, KG, and CAD
- Generate a bar chart comparing all methods

Outputs Generated

1. SFI Values (Per Example)

Stored as lists: - `sfi_values` (ToT) - `sfi_rag` - `sfi_kg` - `sfi_cad`

2. Average SFI per Method

Example printout:

```
Average SFI (ToT): 0.144
Average SFI (RAG): 0.131
Average SFI (KG): 0.202
Average SFI (CAD): 0.005
```

3. Visual Comparison

A bar chart titled:

“Average Self-Focus Index (SFI) Across Mitigation Methods”

Interpretation of Results

- **CAD** shows the lowest SFI → most stable reasoning
- **RAG** and **ToT** show moderate stability
- **KG** has the highest SFI → most susceptible to hallucination drift

Lower SFI = better hallucination mitigation.

Common Issues

Missing or Incorrect Drive Path

Ensure all paths use:

```
/content/drive/MyDrive/YOUR_FOLDER/...
```

Key Errors in JSON Files

Different methods use different AMA keys.
Your notebook already handles both:

```
"final_token_attn"
"mean_attention_final_step"
```

Corrupted or Incomplete Files

If an AMA file fails to load → delete and re-upload.

Summary

This notebook: - Computes SFI for four mitigation techniques
- Measures reasoning stability
- Compares hallucination risk
- Produces numerical + visual results
- Supports your final project evaluation